

The polynomial Taylor tree with spectral truncation (Taylor-tree Stage 3)

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Abstract

Units 230–239 complete Stage 3 of the Taylor-tree programme (Astra #27, gates A–D): the *polynomial* version of `thm:TaylorTree/cor:standardintegralexp`. For polynomial phase ξ and amplitude η (finite monomial lists), $\beta > 0$, $k_i > 0$, on the unit box, the standard integral $Z(N) = \int_{(0,1]^d} \eta u^h e^{-\beta N u^{2k} + \beta \sqrt{N} u^k \xi(u)} du$ satisfies, for every cutoff $L > 0$ and every $N \geq 1$,

$$\left| Z(N) - \sum_{\mu \in \Lambda_L} N^{-\mu} \sum_{j \leq d-1} A_{\mu,j} (\log N)^j \right| \leq C N^{-L} (1 + \log N)^{d-1}$$

(Headline XXV, `taylorTree_cutoff_bound`), with Λ_L the lattice $(2 \prod k_i)^{-1} \mathbb{N}$ below L , an explicit N -free constant C , and coefficients $A_{\mu,j}$ that are cutoff-free absolutely convergent series over phase orders (Headline XXVI packages this as O and o statements, i.e. an asymptotic expansion in the scale $N^{-\mu} (\log N)^j$). The route never estimates a moment at a large exponent: high-spectrum entries are bounded on the τ -side before substitution, phase orders are summed under the absolute integral by a Tonelli identity that folds $\sum_p (\beta B)^p / p!$ into the exponent, and coefficient bounds come from a weighted budget on the state-density lists that is uniform in the monomial. Numerics ($d = 1$, $\xi = 0.3 + 0.5u$, $\eta = 1 + u$, $L = 5/2$): the error times $N^{5/2}$ is ≈ 0.15 for $N = 10, 40, 160, 640$. No `sorry` and no additional `axiom` declarations; 9115 jobs.

1 Architecture

Spectral lattice (u230, `SpectralLattice`). With $Q = 2 \prod_i k_i$ every density exponent ($e + 1)/(2k_i)$ lies in $Q^{-1} \mathbb{N}_{>0}$; distinct lattice points are $\geq 1/Q$ apart; the cutoff set $\Lambda_L = \{m/Q : m < \lceil LQ \rceil\}$ is a `Finset`.

Weighted budget (u231, `DensityBudget`; Gate A). For a representation $v = \sum c \tau^{\mu-1} (-\log \tau)^j$ define $B_Q(v) = \sum |c| j! Q^j$. Through the G_j recursion of the one-coordinate convolution ($|\alpha| \geq 1/Q$ for non-coincident exponents) one gets $B_Q(\text{conv}) \leq (r+1) Q B_Q(v)$ at each step, hence $B_Q(v_w) \leq (n+1)! Q^n$ for every lattice-supported weight vector w on $n+1$ coordinates, *uniformly in the monomial* (`budget_stateDensityRep_1e`). Consequences: $\sum |c| \leq B_Q$ and $|\text{coeffAt}(\mu, j)| j! Q^j \leq B_Q$.

Polynomial ℓ^1 algebra (u233, `MonomialRep`). Lists of (exponent, coefficient) pairs with $\|P\|_1 = \sum |c|$: $|P(u)| \leq \|P\|_1$ on the closed cube, $\|PQ\|_1 = \|P\|_1 \|Q\|_1$, $\|P^p\|_1 \leq \|P\|_1^p$, and the fluctuation part $J = \xi - \xi(0)$ as the list without the constant term.

Log majorant and Tonelli (u234, PhaseMajorant; Gate B). $M_{\nu,r,p}(b) = \int_0^\infty t^{\nu-1}(1 + |\log t|)^r (\sqrt{t})^p e^{-\beta t + \beta b \sqrt{t}} dt$ is finite for $\beta, \nu > 0$ and *every* real b (binomial reduction to the Stage 1 basis on $(0, 1]$, exponential domination on $(1, \infty)$), and

$$\sum_{p \geq 0} \frac{(\beta B)^p}{p!} M_{\nu,r,p}(b) = M_{\nu,r,0}(b + B) \quad (B \geq 0),$$

absolutely convergent with no smallness assumption on B : pointwise $\sum_p (\beta B \sqrt{t})^p / p! = e^{\beta B \sqrt{t}}$ and `hasSum_integral_of_summable_integral_norm` with the partial sums dominated by the folded majorant.

Integrated phase Taylor identity (u235, PhaseTaylorIdentity; Headline XXIV). With $a = \xi(0)$, $J = \xi - a$, $P_p = \eta J^p$, for every fixed $N \geq 0$, $Z(N) = \sum_p \frac{\beta^p}{p!} \int_{(0,1]^d} P_p(u) u^h g_p(N u^{2k}) du$, the interchange justified by the constant bound $\|\eta\|_1 (\beta \|J\|_1 \sqrt{N})^p / p! e^{\beta \sqrt{N} |a|}$ on the box. Each phase order is a *finite* sum of monomial phase integrals, each of which is the exact Stage 2 identity XXIII; the combination (`polyPhaseIntegral_eq_tsum_truncSum`) is the exact polynomial Taylor tree before any truncation (checked numerically to $2 \cdot 10^{-15}$ in $d = 1$).

High spectrum (u236, HighSpectrumBound; Gate C). For an entry (μ, j, c) with $\mu \geq L$, $j \leq n$, $N \geq 1$: $\int_0^1 \tau^{\mu-1} (-\log \tau)^j g_p(N \tau) d\tau \leq N^{-L} (1 + \log N)^n M_{L,n,p}(a)$ (lower the exponent on $(0, 1]$, substitute $t = N \tau$, use $|\log N - \log t| \leq (1 + \log N)(1 + |\log t|)$, enlarge to $(0, \infty)$). Summing with the budget, the ℓ^1 algebra and the Tonelli identity, $|R_{\text{high}}(N)| \leq K_k \|\eta\|_1 (n+1)! Q^n M_{L,n}(a + \|J\|_1) N^{-L} (1 + \log N)^n$, and $Z = \text{lowSeries} + R_{\text{high}}$.

Spectral coefficients (u237, SpectralCoefficients; Gate D). The low part with full moments is regrouped by the *aggregated* coefficient `coeffAt` (never a single list entry) over $\Lambda_L \times \{0, \dots, n\}$ and the binomial sum is reflected, giving $\text{mainPart}_p = \sum_{\mu \in \Lambda_L} \sum_{j \leq n} N^{-\mu} (\log N)^j \text{coeffTerm}_p(\mu, j)$ with

$$\text{coeffTerm}_p(\mu, j) = K_k \sum_{(\gamma, c) \in P_p} c \sum_{q=j}^n \text{coeffAt}(\rho_{h+\gamma}, \mu, q) \binom{q}{j} \int_0^\infty t^{\mu-1} (-\log t)^{q-j} g_p(t) dt.$$

$A_{\mu,j} = \sum_p \frac{\beta^p}{p!} \text{coeffTerm}_p(\mu, j)$ is defined without any cutoff and converges absolutely ($|\text{coeffAt}| \leq (n+1)! Q^n$, $\binom{q}{j} \leq 2^n$, $|\text{fluctMoment}| \leq M_{\mu,n,p}$, Tonelli); at $\mu \leq 0$ all coefficients vanish. Hence $\sum_p \frac{\beta^p}{p!} \text{mainPart}_p = \sum_{\mu \in \Lambda_L} N^{-\mu} \sum_j A_{\mu,j} (\log N)^j$ exactly.

Low-spectrum tail (u238, LowSpectrumTail; Headline XXV). The truncated moments of the low part differ from the full ones by tails $\int_N^\infty$, dominated for $0 < \mu \leq L$ by $\int_N^\infty \log \text{Majorant}_{L,n,p}$; phase orders are summed first (Tonelli on (N, ∞)), the folded tail is $\leq C_1 e^{-\beta N/4}$ and $e^{-\beta N/4} \leq [L]! (4/\beta)^{[L]} N^{-L}$. Assembling: $Z - \sum_{\mu \in \Lambda_L} N^{-\mu} \sum_j A_{\mu,j} (\log N)^j = R_{\text{high}} - \text{tailSeries}$ exactly, and the displayed bound follows with $C = \text{highConst} + \text{tailSeriesConst}$.

Asymptotic packaging (u239, TaylorTreeAsymptotic; Headline XXVI). $Z - \text{spectralSum}_L = O(N^{-L} (1 + \log N)^{d-1}) = O(N^{-L} (\log N)^{d-1})$ and $= o(N^{-L'})$ for every $L' < L$.

2 Correspondence with the paper

The Lean N is the paper's n ; the Lean dimension index is $n = d - 1$; the Lean log degree j is the paper's $j - 1$ and $(\log N)^j$ carries the paper's $m = j + 1$. The lattice $\Lambda_L \subseteq Q^{-1}\mathbb{N}$ is coarser than $\Lambda(h, k)$; the coefficients vanish at exponents not carried by any state density, so the expansion is over $\Lambda(h, k)$ in effect (the identification is not exported). The integrals $\beta^p \int_0^\infty t^{\mu-1} (-\log t)^i g_p$ are the paper's $(-\partial_\mu)^i \partial_a^p S_\mu(a)$ as an interpretation, not a formal derivative theorem. Not formalised: analytic (non-polynomial) ξ, η (Stage 4: a coefficient space with a weighted ℓ^1 norm and summability over monomials), $b \neq 1$.

3 Lessons

Sum phase orders under the absolute integral, never the per-order tail constants (they contain $(\lceil \nu \rceil + i + p)!$); bound high exponents on the τ -side before substituting; regroup by the aggregated coefficient with a list-induction lemma over a finite index set; the $\mu \leq 0$ lattice point needs a vanishing-coefficient case, not an integrability argument. Lean gotchas: `tsum_add` and `sum_le_tsum` are now dot-notation on `Summable`; `summable_of_sum_range_le` needs the bound constant named; `convert ... using 1` on `HasSum` spawns instance goals — rewrite the function with a `funext` equation instead; `first | ring | ...` does not fall through when `ring` degrades to `ring_nf`, so put the contradiction alternatives first.